



Samuel Irungu Kigotho

Data Scientist



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samuelirungu



irungus

About me

I am a T-shaped Data Scientist with deep expertise in Python and R, and broad experience across geospatial data science, machine learning, and environmental analytics. My work focuses on agroforestry systems for climate change mitigation and adaptation, where I apply remote sensing, GIS, allometric modelling, species distribution analysis, and carbon sequestration modelling to generate actionable insights.

I design and implement end-to-end data science pipelines, guided by sound design principles and supported by MLOps practices, to optimize both environmental outcomes and economic benefits for small-scale rural landholders. Alongside my technical expertise, I bring strong communication skills, an open-minded approach, and a commitment to lifelong learning to deliver impactful, climate-smart solutions.

Experience

July 2024 - To Date **Data Scientist** *Strathmore University and ICIPE*

- Designed and implemented end-to-end data collection pipelines for agroforestry research with Strathmore University and ICIPE, capturing multi-source field data (*species, geolocation, imagery, DBH, height, canopy, LULC*) via ODK.
- Led data cleaning, integration, and visualization workflows for large agroforestry datasets, ensuring high-quality inputs for modeling and publications.
- Developed and evaluated carbon sequestration models estimating AGB and BGB using hybrid approaches combining allometric equations and ML models (*SVM, RF, GB, XGBoost, MaxEnt*).
Stack: *Python, SVM, RF, GB, XGBoost, MaxEnt, SHAP.*
- Built species distribution and habitat suitability models, producing spatially explicit maps to guide climate-resilient species-site matching for smallholder farmers.
- Designed and deployed the Agroforestry Smart Decision Support System (ASDSS) web portal, delivering data-driven insights for system design, productivity, and ecological benefits.
Stack: *Python, R, Plumber API, Geospatial Data, ML models.m* [Link to the App](#)
- Applied geospatial analytics and machine learning to translate field and spatial data into actionable insights for climate change mitigation, adaptation, and sustainable rural livelihoods.
- Contributed to peer-reviewed research, integrating field measurements, remote sensing, and ML to advance scalable agroforestry solutions.
[Article.](#)

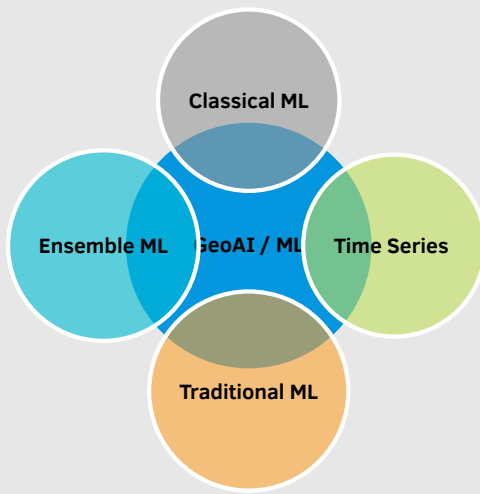
April 2015 - To Date **Senior Research Officer** *Nyandarua County Assembly*

- Providing research and policy analysis services to the Nyandarua County Government and County Assembly, including assessments of policy strengths and weaknesses across key sectors such as Health, Agriculture, Education, Physical Planning, Governance, and Project Monitoring and Management.
- Conducting supporting (auxiliary) research to inform county budgeting processes, guide policy direction, and enhance decision-making for Members of the County Assembly and the general public.
- Offering data analysis and interpretation services to the County Government and Members of the County Assembly to support evidence-based planning and oversight.
- Undertaking budget analysis for various County Government committees throughout the county budgeting cycle to enhance fiscal accountability and efficiency..

2013 - 2015 **Data Processing Manager** *Infotrak Research and Consulting*

- Development of Computer-Aided Personal Interview (CAPI) platforms for efficient data collection using ODK and KoBoCollect.
- Retrieval of survey data from ODK and KoBoCollect servers for processing and analysis.
- Data cleaning and preprocessing to ensure accuracy, consistency, and readiness for analysis.
- Conducting political opinion polls at the national level, including sampling, survey administration, and data collection.
- Data visualization and interactive dashboards using R (including Shiny) and Python for dynamic exploration and monitoring of results.
- Data analysis, reporting, and dissemination of findings to stakeholders, policy-makers, and the public through comprehensive reports and interactive dashboards.

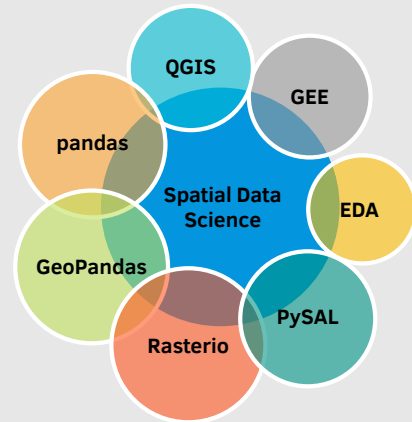
ML Categories



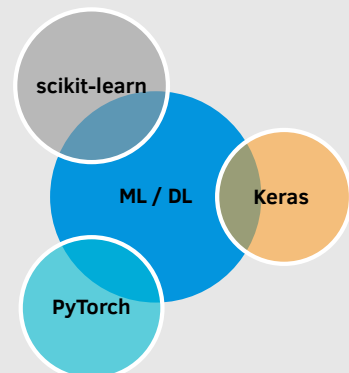
Programming Skills

- Python, R
- Linux • SQL
- MLOps • Git
- AWS • Snowflakes

Spatial Tools



ML Frameworks



Education

2023 May-2026 **MSc in Data Science and Analytics** [Strathmore University](#)

- Developed strong knowledge in applied machine learning, data mining, and predictive analytics to extract insights from structured and unstructured data.
- Developed strong knowledge in building and evaluating statistical and probabilistic models, including linear models, Bayesian methods, and large-scale inference.
- Learned to develop time series forecasting and optimization models to support planning, efficiency, and performance improvement.
- Developed strong knowledge in computational techniques and big data statistical methods for scalable analytics solutions.
- Learned to communicate analytical insights effectively while adhering to ethical data science and responsible AI practices.

2009 - 2013 **BSc in Applied Statistics with Computing** [Maasai Mara University](#)

- Developed strong knowledge in applied statistical inference, regression analysis, hypothesis testing, and estimation theory for data-driven problem solving.
- Developed strong knowledge in probabilistic and stochastic modeling, including probability theory, stochastic processes, survival analysis, and time series analysis.
- Learned to apply multivariate analysis, categorical data analysis, and experimental design to analyze complex real-world datasets.
- Developed strong knowledge in statistical computing, computer programming, databases, and numerical methods to support analytical workflows.
- Built a strong foundation in mathematical statistics, linear algebra, calculus, real analysis, and operations research for advanced analytics and optimization.

Technical Expertise

Expertise Link

- **Spatial Data Science:** Remote Sensing, Geospatial Modelling, Spatial Analysis.
- **Remote Sensing:** Sentinel-1 (SAR), Sentinel-2, Landsat 8/9, Google Earth Engine.
- **Machine Learning:** Regression, Classification RF, GBM, XGBoost, SVM, MaxEnt, Model Optimization, Cross-validation.
- **Deep Learning:** CNNs, LSTMs (basic implementation for image and time series analysis).
- **Time Series Analysis:** Trend detection, anomaly detection, forecasting
- **MLOps:** Pipeline development, API deployment (Flask, Plumber), reproducibility.
- **Programming:** Python, R, SQL

Certificates

[AWS](#) Certified AWS Cloud Practitioner [Azubi Africa AWS/reStart Program](#)

- Built a strong foundation in AWS and services like EC2, S3, IAM, VPC, RDS, ECR, EKS and DynamoDB, and CloudWatch among others.
- Worked with Linux environments and cloud networking to support data processing and analytics workloads.
- Deployed and managed cloud-based data applications, including storage, compute, and monitoring components.
- Applied security, access control, and cost optimization best practices for data platforms on AWS.

Referees

Can be provided on request.